

(Dated: May 2026)

I. QUANTUM TOMOGRAPHY AT WORK

A. One Spin-1/2 system

Let's describe the one spin-1/2 density matrix:

$$\rho = \frac{1}{2} (\mathcal{I} + \vec{B} \cdot \vec{\sigma}) \quad (1)$$

If we want to analyze a decay $\tau/\bar{\tau} \rightarrow l^+/l^-$, then the option of measure just two directions (up/down) isn't available, instead we need to measure in a space, in a solid angle space. So, we have the intention of measure in a specific direction in which the product of the decay goes on, say $\hat{n}$, then the projector operator will take the form:

$$\Pi_{\hat{n}} = \frac{1}{2} (\mathcal{I} + \hat{n} \cdot \vec{\sigma}) \quad (2)$$

The probability that the decay product be emitted in the $\hat{n}$ direction is:

$$\begin{aligned} p(\hat{n}) &= \text{Tr} \left[\frac{1}{2} (\mathcal{I} + \vec{B} \cdot \vec{\sigma}) \frac{1}{2} (\mathcal{I} + \hat{n} \cdot \vec{\sigma}) \right] \\ &= \text{Tr} \left[\frac{1}{4} \left(\mathcal{I} + \vec{B} \cdot \vec{\sigma} + \hat{n} \cdot \vec{\sigma} + \underbrace{(\vec{B} \cdot \vec{\sigma})(\hat{n} \cdot \vec{\sigma})}_{(\vec{B} \cdot \hat{n})I + i(\vec{B} \times \hat{n}) \cdot \vec{\sigma}} \right) \right] \\ &= \frac{1}{4} \text{Tr} \left[\mathcal{I} + \vec{B} \cdot \vec{\sigma} + \hat{n} \cdot \vec{\sigma} + (\vec{B} \cdot \hat{n})I + i(\vec{B} \times \hat{n}) \cdot \vec{\sigma} \right] \\ &= \frac{1}{4} \text{Tr} \left[\mathcal{I} + \left(\vec{B} + \hat{n} + i(\vec{B} \times \hat{n}) \right) \cdot \vec{\sigma} + (\vec{B} \cdot \hat{n})I \right] \\ &= \frac{1}{4} \left[\underbrace{\text{Tr}[\mathcal{I}]}_4 + \underbrace{\text{Tr} \left[\left(\vec{B} + \hat{n} + i(\vec{B} \times \hat{n}) \right) \cdot \vec{\sigma} \right]}_{\vec{\sigma}'\text{'s are traceless}} + \text{Tr} \left[(\vec{B} \cdot \hat{n})I \right] \right] \\ &= \frac{1}{4} [2 + 2\vec{B} \cdot \hat{n}] \end{aligned} \quad (3)$$

So, the probability is given by:

$$p(\hat{n}) = \frac{1}{2} [1 + \vec{B} \cdot \hat{n}] \quad (4)$$

However, the probability isn't normalized in the solid angle space. Let's say that $P(\hat{n}) \propto 1 + \vec{B} \cdot \hat{n}$, then:

$$\int d\Omega p(\hat{n}) = 1 \quad (5)$$

$$\int d\Omega [c(1 + \vec{B} \cdot \hat{n})] = 1 \quad (6)$$

$$\underbrace{c \int_{4\pi} d\Omega}_{4\pi} + \underbrace{c \vec{B} \cdot \int d\Omega \hat{n}}_0 = 1 \quad (7)$$

$$c = \frac{1}{4\pi} \quad (8)$$

Finally, the probability density is given by:

$$p(\hat{n}) = \frac{1}{4\pi} [1 + \vec{B} \cdot \hat{n}] \quad (9)$$

Finally, to perform Quantum Tomography we need to include a *Spin-analyzing power* parameter, κ , in order to cunatify how efficiently the $\hat{n}$ direction tracks the original polarization of the mother particle $\vec{B}$.

$$p(\hat{n}) = \frac{1}{4\pi} \left[1 + \kappa \vec{B} \cdot \hat{n} \right] \quad (10)$$

B. Important step to further probe

An important step that isn't detailed in the literature, and also I still don't completely understand it, is that to obtain a full probability density function that the decay product be projected ($\hat{n}$) in a particular direction (+ or -), is multiply the probabilities of each path; that is:

$$p(\hat{n}^+, \hat{n}^-) = \frac{1}{4\pi} \left[1 + \kappa_+ B_i^+ n_i^+ \right] \frac{1}{4\pi} \left[1 + \kappa_- B_j^- n_j^- \right] \quad (11)$$

$$p(\hat{n}^+, \hat{n}^-) = \frac{1}{(4\pi)^2} \left[1 + \kappa_+ B_i^+ n_i^+ + \kappa_- B_j^- n_j^- + \kappa_+ \kappa_- \sum_{i,j} C_{ij} n_i^+ n_j^- \right] \quad (12)$$

C. Expected values

Since we already have an expression for the probability density function, the next step is try to find an expected value for a particular direction for the decay product.

1. $\langle \hat{n}^+ \rangle$

$$\begin{aligned} \langle n_i^+ \rangle &= \int d\Omega^+ d\Omega^- n_i^+ p(n^+, n^-) \\ &= \int d\Omega^+ d\Omega^- n_i^+ \frac{1}{(4\pi)^2} \left[1 + \kappa_+ B_i^+ n_i^+ + \kappa_- B_j^- n_j^- + \kappa_+ \kappa_- \sum_{i,j} C_{ij} n_i^+ n_j^- \right] \\ &= \frac{1}{(4\pi)^2} \int d\Omega^+ d\Omega^- \left[n_i^+ + \kappa_+ B_i^+ n_i^+ n_i^+ + \kappa_- B_j^- n_i^+ n_j^- + \kappa_+ \kappa_- \sum_{i,j} C_{ij} n_i^+ n_j^- \right] \\ &= \frac{1}{(4\pi)^2} \int d\Omega^+ d\Omega^- \left[\underbrace{n_i^+}_{(1)} + \underbrace{\kappa_+ B_i^+ n_i^+ n_i^+}_{(2)} + \underbrace{\kappa_- B_j^- n_i^+ n_j^-}_{(3)} + \underbrace{\kappa_+ \kappa_- \sum_{i,j} C_{ij} n_i^+ n_j^-}_{(4)} \right] \end{aligned} \quad (13)$$

$$\begin{aligned} (1) : \quad & \int d\Omega^+ d\Omega^- n_i^+ = \int \underbrace{d\Omega^-}_{4\pi} \int \underbrace{d\Omega^+ n_i^+}_0 = 0 \\ (2) : \quad & \int d\Omega^+ d\Omega^- \kappa_+ B_i^+ n_i^+ n_i^+ = \kappa_+ B_i^+ \int \underbrace{d\Omega^-}_{4\pi} \int \underbrace{d\Omega^+ n_i^+ n_i^+}_{\frac{4\pi}{3} \delta_{li}} = \frac{(4\pi)^2}{3} \kappa_+ B_i^+ \\ (3) : \quad & \int d\Omega^+ d\Omega^- \kappa_- B_j^- n_i^+ n_j^- = \kappa_- B_j^- \int \underbrace{d\Omega^+ n_i^+}_0 \int \underbrace{d\Omega^- n_j^-}_0 = 0 \\ (4) : \quad & \int d\Omega^+ d\Omega^- \kappa_+ \kappa_- \sum_{i,j} C_{ij} n_i^+ n_j^- = \kappa_+ \kappa_- \sum_{i,j} C_{ij} \int d\Omega^+ n_i^+ n_i^+ \int \underbrace{d\Omega^- n_j^-}_0 = 0 \end{aligned} \quad (14)$$

So, the expected value for the direction n_i^+ for the decay product is:

$$\langle n_i^+ \rangle = \frac{\kappa_+ B_i^+}{3} \quad (15)$$

or, generally:

$$\langle n_i^\pm \rangle = \frac{\kappa_\pm B_i^\pm}{3} \rightarrow B_i^\pm = \frac{3}{\kappa_\pm} \langle n_i^\pm \rangle \quad (16)$$

This $B_i^\pm$ coefficient it's a very usefull for the framework developed in this work. This is due to the contribution to find the density matrix of a two spin-1/2 system. Now, we still need to find an expression for the expected value for the projected direction of the decay product. Since we already have an expression to find it, there is another way to perform the calculation of this projected direction?

$$\langle n_i^+ \rangle = \int d\Omega^+ d\Omega^- n_i^+ \underbrace{p(n^+, n^-)}_{\text{another expression?}} \quad (17)$$

2. Find another $p(n^+, n^-)$

In the particle physics context, the probability of produce a pair $\tau/\bar{\tau}$ and then decay to a product in the $+/-$ direction came given by the cross section for the proces. In this sense,

$$\int d\Omega^+ d\Omega^- \frac{d^2\sigma}{d\Omega^+ d\Omega^-} = \int d\sigma = \sigma_{total} \quad (18)$$

For a probability density function, this result should be normalized:

$$\int d\Omega^+ d\Omega^- \frac{1}{\sigma_{total}} \frac{d^2\sigma}{d\Omega^+ d\Omega^-} = 1 \rightarrow p(n^+, n^-) \propto \frac{1}{\sigma} \frac{d^2\sigma}{d\Omega^+ d\Omega^-} \quad (19)$$

For instance, we can take this expression and perform some computation to get the expected value for a projected particular direction:

$$\begin{aligned} \langle n_i^+ \rangle &= \int d\Omega^+ d\Omega^- n_i^+ \frac{1}{\sigma} \frac{d^2\sigma}{d\Omega^+ d\Omega^-} \\ &= \frac{1}{\sigma} \int d\Omega^+ n_i^+ \frac{d\sigma}{d\Omega^+} \end{aligned} \quad (20)$$

or, generally,

$$\langle n_i^\pm \rangle = \frac{1}{\sigma} \int d\Omega^\pm n_i^\pm \frac{d\sigma}{d\Omega^\pm} \quad (21)$$

Let's replace this expression in [19]:

$$B_i^\pm = \frac{3}{\kappa_\pm} \frac{1}{\sigma} \int d\Omega^\pm n_i^\pm \frac{d\sigma}{d\Omega^\pm} \quad (22)$$

D. $\langle n_i^+ n_j^- \rangle$

Now, we will try to find another observable in this framework, something that can be meassure or can serve in order to obtain a complete description of the density matrix. Let's see the expected value of a product of projections:

$$\begin{aligned} \langle n_i^+ n_j^- \rangle &= \int d\Omega^+ d\Omega^- n_i^+ n_j^- p(n^+, n^-) \\ &= \int d\Omega^+ d\Omega^- n_i^+ n_j^- \left[1 + \kappa_+ B_l^+ n_l^+ + \kappa_- B_m^- n_m^- + \kappa_+ \kappa_- \sum_{l,m} C_{lm} n_l^+ n_m^- \right] \\ &= \int d\Omega^+ d\Omega^- \left[n_i^+ n_j^- + \kappa_+ B_l^+ n_i^+ n_j^- n_l^+ + \kappa_- B_m^- n_i^+ n_j^- n_m^- + \kappa_+ \kappa_- \sum_{l,m} C_{lm} n_i^+ n_j^- n_l^+ n_m^- \right] \end{aligned} \quad (23)$$